

REVIEW

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Unravelling the complexity: a systematic review of complications and implications of adolescent pregnancy in India

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Abstract

Objectives Adolescent pregnancy is an important public health issue in low and middle-income countries. They are vulnerable to pregnancy-related complications with adverse fetal outcomes because of their immature and developing reproductive systems.

Study design This study is a systematic review of peer-reviewed articles focused on maternal and foetal complications related to adolescent pregnancy in India. The review adhered to the PRISMA 2020 guidelines throughout the process.

Methods A comprehensive search of peer-reviewed articles was conducted using PubMed, CINAHL Plus, Scopus, Web of Science, Ovid-Embase, and Medline. Articles were screened based on predefined inclusion and exclusion criteria. Following critical appraisal, thirteen studies were included in the review.

Results The findings reveal a higher incidence of anaemia, hypertensive disorders of pregnancy, preterm delivery, low babies, and NICU admissions among adolescent mothers. The incidence of adolescent pregnancies in India ranges from 6% to 18.3%. Among adolescent pregnancies, the elevated rates of preterm delivery and low birth weight were associated with increased incidence of foetal distress and NICU admission, underscoring the heightened clinical vulnerability of this population.

Conclusions The review concludes that adolescent pregnancy in India has significant negative health impacts on both mothers and babies. Preventative measures should be comprehensive and multi-faceted, including stringent enforcement of legal measures to prevent child marriage, raising awareness among adolescents and communities about pregnancy-related complications, and providing comprehensive, age-appropriate sexuality education.

Keywords Adolescent, Pregnancy, Teenage pregnancy, Maternal health, Sexual and reproductive health

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Introduction

According to the World Health Organization (WHO), adolescence spans from 10 to 19 years of age [1]. This period is marked by rapid physical, psychological, and emotional development as it bridges childhood and adulthood. Adolescent pregnancy is a critical public health issue in both low- and high-income countries.

Globally, the adolescent birth rate (ABR) has declined, but significant variations exist between and within countries. The global ABR has dropped from 64.5 births per 1,000 women (aged 15–19) in 2000 to 41.3 births per 1,000 women in 2023. Despite this decline, the number remains high, with WHO estimating 21 million adolescent pregnancies annually in low-income countries, resulting in about 12 million births [1]. Factors such as child marriage and child sexual abuse contribute to the risk of adolescent pregnancy.

Adolescent pregnancy remains a persistent and complex public health issue in India, with significant implications for the well-being of young mothers and their children. India is home to the world's largest adolescent population, comprising approximately 253 million young people aged 10–19 years. In rural India alone, 8% of women aged 15–19 become pregnant, according to the National Family Health Survey-5, underscoring the urgent need for evidence-based interventions to address adolescent pregnancy in this population [2]. Among married adolescents in this age group, over half (55%) have already started having children. The prevalence of adolescent pregnancy in India is 6.8%, down from 7.9% in NFHS-4 (2015–2016) [3]. However, World Bank data from 2021 shows an adolescent fertility rate of 17 [4].

Numerous risk factors contribute to adolescent pregnancy. In low- and middle-income countries, over 30% of girls marry before age 18, and 14% before age 15 [5]. Factors such as lack of awareness about contraception, low literacy rates, and inadequate access to healthcare contribute to India's high rate of adolescent pregnancy [6]. A systematic review by Chung, Kim, and Lee (2018) found that adolescents with lower socioeconomic status or less education are more likely to become pregnant [7].

Adolescent women, with their developing reproductive systems, are more prone to complications during and after pregnancy, leading to serious health problems and maternal deaths. Adolescent pregnancy significantly impacts both mother and baby, resulting in high mortality and morbidity [8]. It is associated with conditions such as preterm birth, low birth weight, perinatal death, maternal death, and puerperal sepsis [9].

Studies on complications of adolescent pregnancy have shown conflicting results, highlighting the need for a comprehensive systematic review in the Indian context [6, 7]. Several primary research on adolescent pregnancy have been conducted across different regions of India. Understanding and addressing these complications is crucial for advancing public health initiatives, policy reforms, and targeted interventions. This systematic review aims to investigate the various maternal, foetal, and neonatal complications linked to adolescent pregnancy in India.

Methods

Study design

This study is a systematic review of peer-reviewed articles focused on maternal and foetal complications related to adolescent pregnancy in India. The review adhered to the PRISMA 2020 guidelines throughout the process.

Research question

The research question was formulated using the PEO framework (Population, Exposure, and Outcome): "What are the maternal and foetal or neonatal complications in adolescent pregnancy in the Indian population?" Table 1 presents the PEO search tool.

Search strategy

Following the Centre for Reviews and Dissemination 2009 guidelines, an initial search was conducted in the Cochrane Database of Systematic Reviews (CDSR) to establish the need for this review [10]. After confirming no recent systematic reviews, a comprehensive literature search was conducted in databases including PubMed, Ovid-Embase, Medline, CINAHL Plus, Web of Science, and Scopus. Manual searches (reference harvesting) and Google Scholar were also used to identify studies not indexed in databases. The literature search adhered to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA 2020) guidelines (Fig. 1) to identify a range of primary studies [11].

A search strategy was developed using relevant keywords, controlled vocabulary, and MeSH (Medical Subject Headings) terms. The PEO search strategy tool was utilized to identify keywords (Table 1). Boolean operators and conjunctions were used to refine search results. The search query combined terms such as "adolescence" or "teen" or "teenage" and (1) "pregnancy" or "pregnant" or "antenatal" or "antenatal mothers" or "perinatal" or "postnatal" or "post natal mothers" or "adolescent mothers" or "adolescent pregnancy" or "maternal" or "foetal" and (2) "complications" or "outcomes" or "consequences" or "sequelae" and (3) India.

Table 1 PEO Search tool

P-Population	Adolescent girls<19 years of age in India
E-Exposure	Pregnancy
O-Outcome	Complications in mother and infants

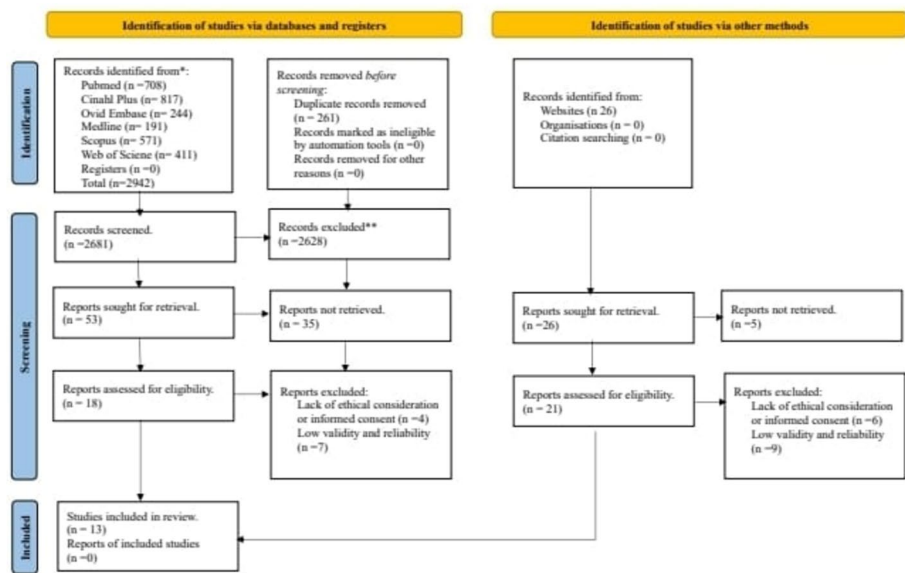


Fig. 1 PRISMA Chart of study selection

Table 2 Inclusion and Exclusion Criteria

Inclusion Criteria	Exclusion Criteria
Adolescent girls < 19 years of age in India	Women aged 20 years and above
Articles reporting complications of adolescent pregnancy	Articles reporting complications other than those of adolescent pregnancy
Peer-reviewed articles	Non-peer-reviewed articles
Published in English language	Not published in the English language
Primary research articles	Reviews, meta-analyses, editorials, commentaries/viewpoints, letters
Quantitative studies	Qualitative studies
Articles published on or after the year 2000	Articles published before the year 2000

Study selection

The inclusion and exclusion criteria are outlined in Table 2. Articles retrieved from databases were imported into RefWorks software, and duplicates were removed. A manual screening was also performed to prevent duplication bias. After removing duplicates, the literature search yielded 259 papers as shown in Fig. 1. This review included only peer-reviewed articles published in English. Two independent reviewers (AVC and RK) conducted all screening phases to ensure consistency and minimize selection bias.

Inclusion and exclusion criteria implementation

After removing duplicates, the literature search yielded 259 papers as shown in Fig. 1. In the first phase, articles were screened for study design. In the next stage, titles and abstracts were screened using the inclusion and exclusion criteria, resulting in 53 articles. Articles lacking

adequate information were excluded, and 35 articles could not be obtained due to unavailability. Ultimately, 39 articles were chosen for critical appraisal. Disagreements between reviewers at any screening stage were resolved through discussion between the two reviewers, with consultation from a third reviewer (ADP) when consensus could not be achieved.

Critical appraisal

Critical appraisal was conducted on the 39 articles to reduce bias, assess validity, check reliability, and evaluate strengths and weaknesses. Case-control and cohort studies were assessed using the Critical Appraisal Skills Programme (CASP) (Tables 3 and 4) [37]. Cross-sectional studies were evaluated using the Appraisal tool for Cross-sectional Studies (AXIS) (Table 5) [44]. Ethical appraisal was also conducted to enhance the ethical and methodological quality of the review.

Outcome of critical and ethical appraisal

Critical and ethical appraisal were conducted in two sequential phases on the 39 articles retrieved for full-text assessment. In the first phase, ethical appraisal assessed the presence of ethical approval and documented informed consent from study participants. Thirteen articles lacked clear ethical approval and/or documented informed consent procedures and were excluded at this stage, leaving 26 articles for further evaluation. In the second phase, the remaining 26 articles underwent critical appraisal for methodological quality, validity, and relevance using standardized tools (CASP for cohort and case-control studies; AXIS for cross-sectional studies). Thirteen articles were excluded due to low validity and

Table 3 Critical Appraisal for Cohort studies using the Critical Appraisal Skills Programme (CASP) tool

References	Section A: Are the results valid?					Section B: What are the results?				Section C: Will the results help locally?			
	Did the study address a focused issue?	Was the cohort recruited acceptably?	Was the exposure accurately measured to minimise bias?	Was the outcome accurately measured to minimise bias?	Have they taken account of the confounding factors in the design and/or analysis?	Was the follow-up of subjects complete enough?	Was the follow-up of subjects long enough?	What are the results of this study?	How precise are the results?	Do you believe the results?	Can the results be applied to the local population?	Do the results of this study fit with other available evidence?	What are the implications of this study for practice?
Banerjee et al. 2009 [12]	+	-	±	±	±	-	-	+	-	+	±	+	±
Bindal DJ 2018 [13]	+	±	+	+	-	-	-	+	+	+	±	±	±
Yasmin, Kumar & Parihar 2014 [14]	+	+	+	+	-	+	+	+	-	+	±	+	+
Ramachandra C et al 2016 [15]	+	+	+	+	+	±	+	±	-	+	±	±	±
Sonali, Priyanka & Sinha 2020 [16]	+	±	+	+	±	±	+	±	-	±	±	-	±
Trivedi & Pasirija 2007 [17]	+	±	±	+	-	±	+	±	+	±	+	+	+
Singh, Kumari and Sinha 2022 [18]	+	±	+	±	-	±	±	±	-	+	±	±	-
Doddihal and Katti 2017 [19]	+	-	+	+	-	+	±	±	+	+	±	±	-
Thobbi et al. 2017 [20]	+	±	+	+	+	±	±	+	-	+	±	±	-

(+) = item adequately addressed, (±) = item not adequately addressed, (+/-) = item partially addressed

(+) = item adequately addressed, (-) = item not adequately addressed, (+/-) = item partially addressed

relevance, including inadequate methodological quality, poor outcome measurement, or outcomes not directly addressing the research question. The final 13 articles met all criteria for ethical standards, methodological quality, validity, and relevance to the research question and were retained for the systematic review.

Data abstraction & analysis

Data were extracted using Microsoft Excel by two authors, including in-text citation of the article, study design, context, duration, inclusion criteria, sample size, data source, aim, key findings, and limitations. Data from quantitative studies were included in this review. Meta-analysis was not feasible due to the heterogeneity of outcome variables and variations in sample sizes among the articles, so a narrative synthesis was conducted.

Results

Characteristics of included studies

This systematic review included 13 primary research articles conducted in various regions of India. Five articles specifically focused on obstetric complications [9, 21, 24, 40, 41], while the remaining articles assessed both maternal and foetal complications and overall outcomes associated with adolescent pregnancy.

Designs of included studies

The 13 quantitative studies included in this systematic review comprised nine case-control studies, one cohort study, and three cross-sectional studies. One study was community-based [19], while the others were conducted in hospital settings.

Sources of data

Among the 13 articles analysed, five gathered data from hospital records [9, 25–27, 34], three utilized interviews with pre-structured and pretested questionnaires [19, 29, 40], and three employed a combination of both methods [24, 41, 42]. Notably, two articles did not specify their data collection methods [21, 28]. Study characteristics such as study design and methods, context, duration, inclusion criteria, sample size, data sources, aims, key findings, and limitations are presented in Table 6.

Maternal and foetal/neonatal complications

Analysis of the data extracted from the included studies reveals both maternal and foetal/neonatal complications associated with adolescent pregnancy. Each study measured different outcomes. The main themes that emerged during the analysis are maternal complications during the antenatal, intrapartum, and postnatal periods, and foetal or newborn complications.

Table 4 Critical Appraisal for case-control studies using the Critical Appraisal Skills Programme (CASP) Tool

Section A: Are the results of the trial valid?										Section B: What are the results?				Section C: Will the results help locally?	
References	Did the study address a focused issue?	Did the authors use an appropriate method to answer their question?	Were the cases recruited acceptably?	Were the controls selected acceptably?	Was the exposure accurately measured to minimise bias?	Aside from the experimental intervention, were the groups treated equally?	Did the authors take account of the potential confounding factors in the design and/or in their analysis?	How large was the treatment effect?	How precise was the estimate of the treatment effect?	Do you believe the results?	Can the results be applied to the local population?	Do the results of this study fit with other available evidence?			
Dutta and Prashant 2013 [21]	+	+	±	±	±	+	-	+	±	+	+	+	+		
Kumar 2007 [22]	+	+	±	±	+	+	-	+	+	+	±	+	+		
Kumari R, Kumari U and Sinha 2020 [23]	+	+	±	±	+	+	+	±	±	+	±	±	±		
Medhi et al. 2016 [24]	+	+	±	±	+	+	+	±	+	+	±	+	+		
Devi OS et al. 2019 [9]	+	+	±	±	±	+	-	+	+	+	±	+	+		
Rudra, Bal & Singh 2013 [25]	+	+	±	±	+	±	+	+	+	+	±	+	+		
Sagili et al. 2012 [26]	+	+	±	±	±	+	±	+	+	+	+	+	+		
Shruthi, Sheela & Kesani VP 2019 [27]	+	+	±	±	+	+	+	+	+	+	±	+	+		
Raut and Patil 2014 [28]	+	+	±	±	+	+	+	+	+	+	±	+	+		
Seneesh and Shah 2015 [29]	+	+	+	+	+	+	+	+	+	+	+	+	+		
Sudhir et al. 2015 [30]	+	+	±	±	-	+	+	-	-	±	-	+	+		
Malviya 2003 [31]	+	+	±	±	±	+	±	+	±	+	±	±	±		
Kumari U 2019 [32]	+	+	+	+	+	±	+	±	+	+	±	+	+		
Patidhar 2015 [33]	+	+	-	-	+	+	-	±	+	+	-	+	+		
Nair A et al. 2015 [34]	+	+	+	+	+	+	+	+	+	+	+	+	+		
Devi G et al. 2019 [35]	+	+	+	-	+	NA	-	+	-	+	±	±	±		
Thaker et al. 2013 [36]	+	+	+	-	+	NA	-	±	±	±	±	±	±		

(+) = item adequately addressed, (-) = item not adequately addressed, (+/-) = item partially addressed, NA = Not Applicable

Maternal and foetal/neonatal complications

The incidence of adolescent pregnancy in the included studies ranged from 6% to 18.3% [25–27, 34, 41, 42]. Analysis of the data extracted from the included studies reveals both maternal and foetal/neonatal complications associated with adolescent pregnancy. These complications are organized by timing of occurrence and categorized as follows.

I. MATERNAL COMPLICATIONS

Adolescent mothers experienced various maternal complications throughout pregnancy and the postpartum period. These complications are categorized by timing of occurrence as antenatal (during pregnancy), intrapartum (during labour and delivery), and postnatal (after delivery) complications.

A. Antenatal Complications

The majority of adolescent mothers experienced antenatal complications. Commonly reported antenatal complications included anaemia, hypertensive disorders of pregnancy (including pregnancy-induced hypertension and pre-eclampsia), oligohydramnios, gestational diabetes mellitus, preterm labour, and post-datism.

Incidence rates for antenatal complications were substantial across studies. Dutta and Prasant (2013), Raut and Patil (2014), and Mohite et al. (2014) reported antenatal complications incidence rates of 68.8%, 65.3%, and 48.38%, respectively [21, 28, 41], demonstrating the high prevalence of antenatal complications in adolescent pregnancies.

Anaemia was the most frequently reported antenatal complication in the reviewed articles, with incidence rates ranging from 44% to 81.7% across different studies. Most studies concluded that anaemia was significantly associated with adolescent pregnancy, except Sagili et al. (2012), where anaemia was significantly higher in the non-adolescent age group [26]. Severe anaemia incidence in adolescent pregnancy was reported by Seneesh & Shah (2015) and Doddihal & Katti (2017) at 11.4% and 4.1%, respectively [19, 29].

Hypertensive disorders of pregnancy, including Pregnancy Induced Hypertension (PIH) [28, 29, 41, 42] and Pre-eclampsia [24, 34], were significantly associated with adolescent pregnancy and were reported in multiple studies. Although eclampsia was observed in four studies, it was not found to be statistically significant [24, 27, 29, 34].

Oligohydramnios was related to adolescent pregnancy in three studies [9, 21, 41]. None of the studies found a link to gestational diabetes mellitus, antepartum haemorrhage, or maternal mortality.

B. Intrapartum Complications

Intrapartum complications were documented in several studies examining adolescent pregnancies. Specific intrapartum complications included non-progression of labour, deep transverse arrest [28], perineal tears, and episiotomy [26], indicating increased obstetric complexity during labour and delivery.

The majority of adolescent mothers underwent normal vaginal delivery, which was statistically significant [9, 21, 25, 26, 40–42], while the rest underwent either instrumental delivery or caesarean section. Only Rudra, Bal, & Paul (2013) described emergency caesarean sections, which were not statistically significant [25].

Although many authors describe preterm delivery as common in adolescent pregnancy, three studies pointed to an association with post-datism (prolonged pregnancy) [26, 28, 40]. Two articles mentioned significantly higher rates of premature rupture of membranes (PROM) [25, 42], while Sagili et al. found PROM significantly higher in the non-adolescent age group [26]. Sagili et al. also did not find a significant relationship with preterm delivery in adolescence. However, it should be noted that not all included studies described intrapartum complications comprehensively, limiting the completeness of data on this complication category.

C. Postnatal Complications

Postpartum complications were described in three studies included in this review. Postpartum haemorrhage was the primary postpartum complication documented. However, postpartum haemorrhage was not found to be significantly associated with adolescent pregnancy in the studies reviewed, suggesting that postpartum bleeding complications may not be uniquely elevated in this population compared to adult pregnancies.

II. FOETAL/NEONATAL COMPLICATION

Complications in the newborn included low birth weight, respiratory distress, and subsequent NICU admission. Low birth weight was noted in the majority of studies on adolescent pregnancy, with two studies reporting significant incidences of very low birth weight babies [27].

Preterm delivery resulted in low-birth-weight babies among adolescent women, contributing to higher NICU admission rates. Five studies demonstrated birth asphyxia [21, 28, 40–42].

Intrauterine growth retardation was more frequent in adolescent pregnancies compared to the non-adolescent age group [21, 25, 28, 41]. Of the six articles mentioning NICU admission, five concluded it was significantly

Table 5 Critical appraisal for cross-sectional studies using the Appraisal Tool for Cross-Sectional Studies (AXIS)

Articles	Introduction	Methods											
		Were the aims/objectives of the study clear?	Was the study design appropriate for the stated aim(s)?	Was the sample size justified?	Was the target/reference population clearly defined? (Is it clear who the research was about?)	Was the sample frame taken from an appropriate population base so that it closely represented the target/reference population under investigation?	Was the selection process likely to select subjects/participants that were representative of the target/reference population under investigation?	Were measures undertaken to address and categorise non-responders?	Were the risk factors and outcome variables measured appropriate to the aims of the study?	Were the risk factor and outcome variables measured correctly using instruments/measurements that had been trialled, piloted, or published previously?	Is it clear what was used to (in-)clude/determine statistical significance and/or sufficiently precisely described to enable them to be repeated?	Were the risk factor and outcome variables measured correctly using instruments/measurements that had been trialled, piloted, or published previously?	Is it clear what was used to (in-)clude/determine statistical significance and/or sufficiently precisely described to enable them to be repeated?
Borah and Das 2022 [38]	+	+	-	+	+	+		NS	+	+	+	-	
Kesava Chandra et al. 2022 [39]	+	+	-	+	+	+		-	±	-	-	-	
Mukhopadhyay, Chaudhuri &Paul 2010 [40]	+	+	+	+	+	±		NS	+	+	+	+	
Mohite et al. 2014 [41]	+	+	-	+	+	+		NS	+	±	+	±	
Anand et al. 2023 [42]	+	+	+	+	+	+		NS	+	+	+	+	
Jain and Gupta 2019 [43]	+	+	±	+	+	+		NS	+	±	+	+	
Articles	Results			Discussion				Others					
	Were the basic data adequately described?	Does the response rate raise concerns about non-response bias?	If appropriate, was information about non-responders described?	Were the results internally consistent?	Were the results of the analyses described in the methods, presented?	Were the authors' discussions and conclusions justified by the results?	Were the limitations of the study discussed?	Were there any funding sources or conflicts of interest that may affect the authors' interpretation of the results?	Was ethical approval or consent of participants attained?				
Borah and Das 2022 [38]	+	NS	-	±	+	+	-	+	+				
Kesava Chandra et al. 2022 [39]	±	NS	-	-	±	+	+	-	+				
Mukhopadhyay, Chaudhuri &Paul 2010 [40]	+	NS	-	+	+	+	+	-	+				
Mohite et al. 2014 [41]	+	NS	NA	+	+	+	-	-	+				
Anand et al. 2023 [42]	+	NS	NA	+	+	+	+	-	+				
Jain and Gupta 2019 [43]	+	NS	NA	+	+	+	-	-	-				

associated with babies of adolescent mothers compared to non-adolescent mothers [9, 24, 25, 27, 42].

Two studies observed higher rates of stillbirths in babies of adolescent mothers compared to non-adolescent mothers [26, 40]. Raut and Patil (2014) observed significantly higher early neonatal deaths in the adolescent age group, accounting for 83.3% of all neonatal deaths [28].

Discussion

This systematic review was conducted to identify the impact of adolescent pregnancy on the health of the mother and the baby. The review indicated that adolescent women had an elevated risk for complications during pregnancy, including both systemic and obstetric consequences. While the adolescent pregnancy rate has decreased in developed countries, it persists as a public health issue in developing countries. In 2021, approximately 14% of adolescent pregnancies were present globally, with 10% occurring in South Asia [45].

In India, the wide variation in adolescent pregnancy outcomes and reproductive health status across regions is attributable to significant disparities in healthcare infrastructure, availability of reproductive health services, and implementation of adolescent-friendly health programs. Stillbirth and unsafe abortion are particularly concerning complications in adolescent pregnancies. Adolescents face higher risks for unsafe abortion compared to adult women due to limited access to safe reproductive health services, inadequate reproductive health knowledge, and significant social and economic barriers [23]. These adverse outcomes can affect adolescent mothers physically and emotionally during their critical developmental transition from childhood to adulthood, with potential for long-term health and psychological consequences. The adverse outcomes of adolescent pregnancy in India can also be attributed to low socioeconomic backgrounds and child marriage. According to a study conducted by Shri et al. (2023) in Uttar Pradesh and Bihar, 61% of married adolescent women became pregnant before age 20, with 18.57% experiencing miscarriage, abortion, or stillbirth [46]. Adolescent pregnancy is influenced by early marriage and sexual abuse. Pregnant adolescents are less likely to seek proper antenatal care, which contributes to complications in this age group [47]. Low socioeconomic status, large family size, and higher birth order also affect the outcomes of adolescent pregnancy. A multi-country study in Sub-Saharan Africa by Mombo-Ngombo et al. (2016) categorizes adolescent mothers into a high-risk group, suggesting that routine care may not be sufficient to avoid complications [48].

Differences in the studies can be attributed to the services mothers receive, study area (urban or rural), and study duration [41]. From the thirteen included

studies, anaemia, preterm delivery, and low birth weight were consistent complications associated with adolescent pregnancy. Devi et al. (2019) found that 86% of the adolescent age group were primigravida and that anaemia was not due to young age but nutritional deficiencies, malaria, and hookworm infestation [9]. Adam and Ali (2016) commented that anaemia in pregnancy is due to physiological changes leading to haemodilution and increased iron requirements [49]. Sagili et al. (2012) found that anaemia was significantly higher in non-adolescents, likely due to early iron supplement intake in booked cases [26]. Although hypertensive disorders of pregnancy were significantly associated with adolescent pregnancy, no study found a relation with eclampsia. Only Devi et al. (2019) demonstrated significant oligohydramnios in adolescent mothers, attributing the increased caesarean section rate to immature birth canals and pelvic bones [9]. Preterm delivery resulted in low-birth-weight babies among adolescent women, contributing to higher NICU admission rates. Pre-eclampsia, anaemia, and poor nutritional status are the main causes of premature and low-birth-weight newborns [21]. Raut and Patil (2014) conducted a prospective case-control study comparing 160 adolescent mothers (aged 13–19 years, mean age 15.8 years) with 160 adult control mothers (aged 20–25 years). The study revealed that adolescents experienced significantly higher rates of antenatal complications (65.3% vs. 34.7% in the adult group). Specifically, adolescents demonstrated substantially elevated rates of cephalopelvic disproportion (CPD) (100% vs. 0% in adults), intrauterine growth retardation (IUGR) (83.3% vs. 16.7% in adults), and anaemia (75% vs. 25% in adults). Additionally, adverse foetal outcomes were markedly elevated in the adolescent group, with 70% of caesarean sections, 76% of preterm deliveries, and 56% of low birth weight babies occurring in the adolescent mothers compared to 30%, 24%, and 44% respectively in the adult control group [28]. There are marked variations in reporting caesarean and vaginal delivery rates among adolescents. Raut and Patil (2014) also concluded that CPD contributed to increased operative interventions (70.8%) and instrumental deliveries (61%) [28]. In contrast, Medhi et al. (2016) and Seneesh and Shah (2015) found no increase in caesarean sections among adolescents but reported increased vaginal deliveries, which can be explained by the low birth weight and preterm delivery of the babies [24, 29]. Most studies concluded that preterm delivery was more frequent in adolescent mothers, though Sagili et al. (2012) reported significant post-datism in this age group [26]. Mukhopadhyay, Chaudhuri, and Paul (2010) found that early pregnancy registration and iron and folic acid tablet consumption were lower in adolescent pregnancies, with a higher proportion of stillbirths [40]. Adolescent pregnancy was not

Table 6 Data extraction table: characteristics of included studies and summary of key findings

References	Study Design and Methods	Context	Duration	Inclusion Criteria	Sample size	Source of data	Aim of the study	Key findings	Limitations
Devi et al. 2019 [9]	Retrospective observational study	Mediciti Institute of Medical Sciences, Telangana	July 2015–December 2017	cases: <19 years controls: 20–35 years	cases: 356 controls: 356	Hospital records	To determine whether teenage pregnancies are associated with adverse maternal and perinatal outcome	The prevalence of teenage pregnancy was 7%. Anaemia was the most common complication (44%) followed by PIH (9%). Preterm rate of 8% in teenage pregnancy. High NICU admission and high LBW in newborns of teenage mothers were noted in the current study.	Inability to generalize the findings to the general population as it is a hospital-based study. *
Dutta & Prasant 2013 [21]	Prospective case-control study	Rural Medical Research Centre, South India	November 2010–October 2012	cases: <19 years controls: 20–29 years	cases: 80 controls: 160	not mentioned	To evaluate the maternal and foetal outcomes and complications in teenage primi-gravida as compared to those aged 20–29 years	Anaemia (68.75%) was the most common complication in adolescent pregnancy followed by hypertensive disorders of pregnancy. The incidence of antenatal complications in adolescence was 68.8%. 31.7% of neonates of adolescent mothers required NICU admission. Postnatal complications such as wound gaping (12.8%) and sepsis (5%) were present.	Since it is a hospital-based study, it may not apply to the entire population. *

Table 6 (continued)

References	Study Design and Methods	Context	Duration	Inclusion Criteria	Sample size	Source of data	Aim of the study	Key findings	Limitations
Medhi et al. 2016 [24]	Prospective case-control study	Tertiary Care Hospital, Northeast India	January 2014-December 2014	cases:<19 yearscontrols: 20–25 years	cases: 165 controls: 330	Structured and pretested proforma, interview, hospital records	To find out the obstetric complications during the antepartum, intrapartum, and postpartum period and the perinatal outcome of adolescent primigravidae in India	Complications such as pre-eclampsia (11.52%), Preterm delivery (23.64%), low birth weight (26.06%) and NICU admission (16.36%) were higher in cases. Severe anaemia, eclampsia, mode of delivery, antepartum and post-partum haemorrhage were equal in both cases and controls.	The study was conducted in a tertiary hospital, where high-quality care facilities are available hence cannot be generalised to the whole population. Adolescents <15 years were not included.
Nair & Devi 2015 [34]	Retrospective case-control study	Tertiary Care Teaching Hospital, North Kerala	January 2013-June 2013	cases: <19 yearscontrols: 20–29 years	cases: 538 controls: 538	Hospital records	To assess the obstetric outcome of teenage pregnancy and to compare it with that of the control group.	Teenage pregnancy rate was 8.48%. The most common complications in both cases and controls were anaemia followed by preeclampsia. The relative risk of anaemia in the adolescent age group was 1.97 and for pre-eclampsia was 2.76 which is highly significant. 29.49% of newborns had low birth weight with a relative risk of 1.57	Study conducted in a tertiary care hospital hence not generalisable to population*

Table 6 (continued)

References	Study Design and Methods	Context	Duration	Inclusion Criteria	Sample size	Source of data	Aim of the study	Key findings	Limitations
Raut and Patil, 2014 [28]	Prospective case-control study	Tertiary care hospital	2008	cases: 13–19 years controls: 20–25 years	Cases: 120 controls: 120	not mentioned	To study the maternal and foetal outcome of adolescent pregnancy (13–19yrs) and compare the same with pregnancies from the control group (20–25)	65.3% of cases had antenatal complications and 66% had intrapartum complications. Common complications were IUGR and anaemia. The most common intrapartum complications were deep transverse arrest, non-progression of labour and foetal distress. Incidence of LSCS, preterm delivery, low birth weight, birth asphyxia and Neonatal death were higher in cases.	Not generalisable to the entire population*
Rudra, Bal & Singh 2013 [25]	Retrospective case-control study	Tertiary care urban hospital	September 2009 - August 2012	cases: <19 years controls: 20–30 years	Cases: 340 controls: 680	Hospital records	To evaluate the maternal and fetal outcome of teenage pregnancy and find the effect of antenatal care on these teen mothers, to find out the incidence of teenage pregnancy, to compare the outcome of teenage pregnancy with the control group,	The incidence of adolescent pregnancy was 4.33%. Anaemia (9.7%), hypertensive disorders of pregnancy (9.4%), IUGR (6.7%), preterm (11.2%), PROM (3.2%), low birth weight (41.2%) and NICU admission (5.8%) were higher in cases. Vaginal delivery was higher in the control group (82.7%).	Findings are specific to the hospital and may not apply to the population. *

Table 6 (continued)

References	Study Design and Methods	Context	Duration	Inclusion Criteria	Sample size	Source of data	Aim of the study	Key findings	Limitations
Sagili et al., 2012 [26]	Retrospective study	JIPMER, South India	March 2008-April 2009	cases: <19 years controls: >19 years	cases: 620 controls: 14,878	Hospital records	To compare obstetric and perinatal outcomes in teenage and non-teenage pregnancies	The incidence of adolescent pregnancy was 4.17%. The most common complication was postdatism. (26.9%). Adolescent pregnancy had a higher incidence of Normal vaginal delivery (81.5%). Episiotomy was needed more in the study group but there were fewer perineal tears. Low birth weight was significantly higher in cases (33.1%). Anaemia, PROM, and caesarean section were more common in the control group.	Few results were contradictory to the usually held knowledge, probably because of the high-quality care provided in this hospital. *
Seneesh & Shah, 2015 [29]	Prospective case-control study	Vadodara, Gujarat	May 2008-April 2009	cases: <19 years controls: >20 years	cases: 70 controls: 70	Semi-structured open-ended questionnaire, interviews	To study the epidemiological aspects and clinical fetal-maternal outcome of teenage pregnancy	Socioepidemiological variables such as illiteracy, rural background, family size greater than 4, birth order >3, and low socioeconomic status were associated with cases. Mild (85.8%) and severe (11.4%) anaemia, severe PIH (17.1%), Preterm delivery (17.1%) and Low birth weight (77.2%) were higher.	Lack of large sample size due to strict inclusion criteria. Institutional-based study underestimates the magnitude of the problem. Confounding bias might be present. Have not considered postnatal complications.

Table 6 (continued)

References	Study Design and Methods	Context	Duration	Inclusion Criteria	Sample size	Source of data	Aim of the study	Key findings	Limitations
Shruthi, Sheela & Kesani, 2019 [27]	Retrospective case-control study	Tertiary care teaching hospital, rural India	April 2017-March 2018	cases: 15–19 years controls: 20–25 years	cases: 450 controls: 460	Hospital records	To discover the obstetric complexities and the perinatal result of adolescent primigravidae in India	The incidence of adolescent pregnancy was 18.3%. Preeclampsia (12.8%), eclampsia (3.5%), preterm (18.6%), low birth weight (39.1%), extremely low birth weight (13.7%) and NICU admission (20.8%) were higher in cases.	Conducted in a tertiary care centre hence may not apply to the entire population.
Doddihal & Katti 2017 [19]	Prospective study community-based	Jawahar Lal Nehru Medical College, Belgaum	January 2012-December 2012	cases: 15–19 years	cases: 144	Interview with a predesigned pretested questionnaire	To study the complications and socio-demographic profile of adolescent pregnancies	anaemia is associated with the age of the patient. As age increases anaemia decreases. There is a significant association between anaemia and adolescent pregnancy.	Did not consider age < 15 years. And postpartum complications*. Findings may not apply to the community. *
Anand et al., 2023 [42]	Cross-sectional study	Tertiary Care Hospital, Tamil Nadu	January 2021-June 2021	Group 1: < 19 years- Group 2: 20–29	cases: 602	Structured proforma, case records	To estimate the prevalence of adolescent pregnancies in an urban tertiary care hospital and to analyse the socio-demographic factors and the neonatal outcomes associated with them.	The prevalence of adolescent pregnancy is 10.26%. anaemia (56.31%), PIH (4.31%) is highly significant in the group 1. 11.96% of adolescent mothers had PROM. Preterm (30.15%), Intra-uterine death (2.5%), Low birth weight (27%), birth asphyxia (3.24%), and neonatal deaths (2.21%) are higher in group 1	Since this is a hospital-based study, the prevalence in this study may not reflect the prevalence in the community

Table 6 (continued)

References	Study Design and Methods	Context	Duration	Inclusion Criteria	Sample size	Source of data	Aim of the study	Key findings	Limitations
Mohite et al., 2014 [41]	Cross-sectional study	Krishna Hospital, Karad	First quarter of 2012	Group1: <19 years Group 2: 20–30	teenage: 93 non-teenage: 93	Pretested structured proforma by personal interview method and case paper records	To assess the magnitude and association of obstetric profile of teenage and non-teenage primi mothers.	Among the adolescents' the proportion of obstetric complications such as anaemia (81.7%), PIH, oligohydramnios, and fetal distress was high compared to non-adolescents.	Have not considered postpartum complications
Mukhopadhyay, Chaudhuri & Paul, 2010 [40]	Cross-sectional study, observational	RG Kar Medical College and Hospital, Kolkata	June 2006–May 2007	Cases: 13–19 years Comparisons: 20–29 years	cases: 350 comparisons: 350	Interviews by the principal investigator by visiting labour wards, operation theatre	To compare the sociodemographic characteristics and perinatal outcomes of teenage primi with those of adult primi	Adolescent mothers had a higher proportion of preterm deliveries (27.7%). Still birth rate was higher in cases. Adolescent pregnancy was significantly associated with low birth weight and birth asphyxia.	Since it is a hospital-based study chances of high-risk cases are higher and may not reflect the situation in a community setting. Confounding bias with sociodemographic characteristics.

associated with gestational diabetes mellitus, malpresentation, intrauterine death, congenital anomalies, molar pregnancy, antepartum or postpartum haemorrhage, perinatal or maternal mortality, or thyroid disorders in any of the studies included in the review.

Since most studies were conducted in hospital settings, the outcomes might differ for the general population. It was identified that complications in this age group could be significantly reduced where adequate antenatal care and regular check-ups are provided. As this review highlighted, pregnancy can adversely affect the health of adolescents. In the reviewed articles describing anaemia, the type of anaemia was not specified. Preterm delivery of newborns has consequences such as IUGR, birth asphyxia due to respiratory distress syndrome, and retinopathy of prematurity.

This review has some strengths and limitations. It identified adverse foetal, neonatal, and maternal outcomes of adolescent pregnancy in the Indian population through a comprehensive search using PRISMA 2020 guidelines, helping reduce bias from multiple individual studies and providing evidence-based data for more reliable and accurate conclusions. The review identified gaps in existing literature. However, it only included peer-reviewed articles, which is both a strength and a limitation. Additionally, articles where full text could not be retrieved

were excluded, potentially omitting up-to-date research. The review was conducted by a single author without cross-checking, which might have affected the quality of reporting and analysis. This review also did not consider the socio-demographic background of adolescents, leading to potential confounding bias. Some included studies had small sample sizes. Meta-analysis was not performed due to a lack of reported estimated effect sizes, limiting the review's statistical power. This review did not explore the role of dietary habits, intimate partner violence, and socioeconomic profile in leading to complications of adolescent pregnancy.

Conclusions

In conclusion, adolescent pregnancy in India represents a complex public health challenge with significant consequences for maternal and neonatal health, as well as broader societal implications. Pregnancy-related complications, including anaemia and hypertensive disorders, contribute to increased maternal morbidity and may disrupt educational attainment and economic opportunities, thereby perpetuating cycles of poverty. Preterm birth and low birth weight in infants born to adolescent mothers result in higher neonatal morbidity requiring intensive care, with potential long-term developmental and health sequelae. These complications collectively

impose substantial economic burdens on healthcare systems and families.

Multi-level interventions are necessary to address adolescent pregnancy effectively. At the policy level, strengthening and enforcing legislation to prevent child marriage is critical, complemented by development and scaling of adolescent-friendly health services. Prevention-focused strategies should prioritize comprehensive, age-appropriate sexuality education integrated into school curricula, coupled with improved access to contraceptive information, counselling, and methods through healthcare facilities. Healthcare systems must ensure accessible, quality antenatal care including early pregnancy registration, regular monitoring, nutritional support, and targeted counselling for pregnant adolescents. Public health campaigns, tailored to both rural and urban contexts, should communicate the documented health risks and complications associated with adolescent pregnancy, engaging schools, healthcare facilities, community leaders, and other stakeholders as implementation partners. Psychosocial support systems should be strengthened to assist both adolescent mothers and their families in managing the physical, emotional, and social challenges associated with early parenthood.

Addressing adolescent pregnancy requires coordinated action across health, education, social protection, and policy sectors. Given the documented burden of maternal and neonatal complications, reducing adolescent pregnancy should remain a priority area in India's public health agenda, supported by adequate resources, trained personnel, and systematic monitoring of implementation outcomes.

Abbreviations

ABR	Adolescent Birth Rate
CASP	Critical Appraisal Skills Programme
CDSR	Cochrane Database of Systematic Reviews
CRD	Centre for Reviews and Dissemination
MeSH	Medical Subject Headings
NICU	Neonatal Intensive Care Unit
PEO	Population, Exposure, Outcome
PIH	Pregnancy Induced Hypertension
PPH	Postpartum haemorrhage
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
PROM	Premature rupture of membrane
WHO	World Health Organisation
UNICEF	United Nations Children's Fund

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12978-025-02265-x>.

Supplementary Material 1

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During the preparation of this work the authors used ChatGPT to improve language and readability. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Authors' contributions

Anjini Vijayan Clitus: Conceptualization, Methodology, Investigation, Formal analysis, Writing – original draft, Writing – review & editing, Visualization
Ali David Parsa: Conceptualization, Methodology, Investigation, Writing – review & editing, Project administration, Resources, Correspondence Ibrahim Alasqah: Conceptualization, Methodology, Investigation, Writing – review & editing, Project administration, Resources, Correspondence Nada Alqarawi: Methodology, Formal analysis, Writing – review & editing, Validation, Supervision Ilias Mahmud: Methodology, Formal analysis, Writing – review & editing, Validation, Supervision Russell Kabir: Conceptualization, Methodology, Writing – review & editing, Supervision, Project administration, Funding acquisition.

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Data availability

No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate

This study is a systematic review of published research; therefore, ethical approval is not required.

Competing interests

The authors declare no competing interests.

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